

U.S. Multi Industry & Electrical Equipment

Orbital Data Centers: Content (\$M/MW) Implications for Cooling Equipment Players



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As AI infrastructure continues to scale, industry participants are increasingly evaluating alternative approaches to meeting the growing requirements for power, cooling and physical infrastructure. Bernstein recently [initiated coverage on SpaceX](#), highlighting the company's position at the center of the rapidly evolving space economy and its role in enabling new classes of orbital data centers. Building on that work, we explore one of the more technical themes emerging at this intersection of AI and space: how liquid cooling systems would *evolve* in the context of data centers in space. Our hypothesis suggests that the underlying cooling architecture is largely an extension of technologies already used widely in traditional satellite and spacecraft designs. The key question is therefore not whether an orbital data center can be cooled, but rather how the cooling system would be designed, what components would be required, and which suppliers are potentially exposed to the opportunity.

Similar to terrestrial (land-based) facilities, virtually all electrical power consumed by compute hardware ultimately becomes heat. The difference is that, in orbit, heat cannot be removed through convection because no atmosphere exists. Instead, thermal energy must be **rejected through infrared radiation, emitted into deep space using “radiator wings”**. From a liquid cooling perspective, we hypothesize the architecture remains relatively simple. Heat is captured at the chip level using direct-to-chip cold plates and transported through a liquid loop into an anhydrous coolant-based secondary loop connected to deployable radiator wings. In our view, **orbital data center cooling therefore represents an evolution of today's liquid-cooling ecosystem rather than a fundamentally new paradigm**. Traditional facility-level equipment such as chillers, CRAHs, cooling towers and dry coolers would be largely eliminated within an orbital liquid cooling setup. However, components such as cold plates, manifolds and CDUs remain necessary, while an entirely new category emerges in the form of radiator wing systems and associated spacecraft thermal hardware.

Based on our analysis, we estimate the total cooling equipment content opportunity at **\$0.6–0.8M/MW for orbital data centers (excluding radiator wings), versus \$1.3–1.8M/MW for terrestrial liquid-cooled data centers**. Service will look materially different, with **orbital data centers likely seeing minimal (possibly even zero) service attach**. Note, this is assuming terrestrial \$/MW for equipment in space; actual numbers could vary based on final form factors. Market share of this TAM will also likely look different, **especially given the bias of SPCX towards vertical integration putting pressure on incumbents**.

However, this is not a near-term headwind. Based on our [initiation document](#) we estimate that **orbital capacity scales to ~11% of total installed global data center capacity through 2031 (although SPCX's own numbers are higher)**. While we believe the concept is technically credible, **we do not currently view it as a material threat to terrestrial cooling infrastructure demand over the medium-term (although for the long-term, this is very much open to debate)**.

BERNSTEIN TICKER TABLE

Ticker	Rating	31 Jul 2026		Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
VRT (Vertiv)	O	USD	241.57	368.00	47.8%	USD	4.20	6.60	9.06	57.6	36.6	26.7
NVT (nVent)	O	USD	153.83	220.00	78.0%	USD	3.34	4.84	6.26	46.0	31.8	24.6
TT (Trane)	O	USD	454.95	555.00	(14.3)%	USD	13.06	14.98	17.82	34.8	30.4	25.5
JCI (Johnson Controls)	O	USD	146.66	173.00	21.5%	USD	3.78	4.97	5.93	38.8	29.5	24.7
CARR (Carrier)	M	USD	61.81	78.00	(28.1)%	USD	2.57	2.85	3.29	24.0	21.7	18.8
SU.FP (Schneider)	O	EUR	289.60	330.00	5.2%	EUR	8.43	10.42	12.80	34.4	27.8	22.6
LR.FP (Legrand)	O	EUR	131.40	170.00	(20.5)%	EUR	5.22	6.01	6.80	25.2	21.9	19.3
ETN (Eaton)	O	USD	415.20	534.00	(10.2)%	USD	12.07	13.45	16.37	34.4	30.9	25.4
NVDA (NVIDIA)	O	USD	200.75	315.00	(5.3)%	USD	4.77	9.19	12.52	42.1	21.9	16.0
SPCX (SpaceX)	O	USD	108.37	239.00	NA	USD		(0.18)	0.83		(587.4)	131.3
SPX			7,489.72									
EDME			1,611.67									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

NVDA base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

US Multi-Industrials: We rate VRT (PT \$368), NVT (PT \$220), TT (PT \$555) and JCI (PT \$173) Outperform. We rate CARR (PT \$78) Market-Perform.

European Capital Goods: We rate Schneider (PT €330) and Legrand (PT €170) Outperform.

NVDA (Outperform, \$315 PT): The datacenter opportunity is enormous, and still early, with material upside still possible.

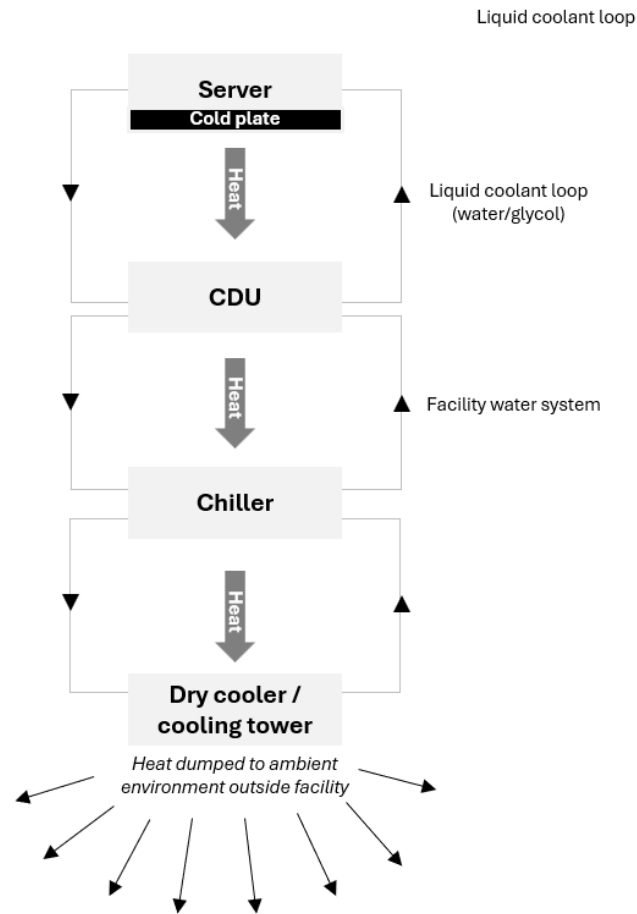
US Machinery & Electrical Equipment: We rate ETN (PT \$534) Outperform

SPCX (Outperform, \$239 TP): We rate SPCX Outperform, TP \$239, using sum of the parts on 2031 EBITDA for each segment, assuming a one-year delay on orbital data center launches. We use late stage venture discount rates, 25% and 35%, respectively, for Connectivity and AI.

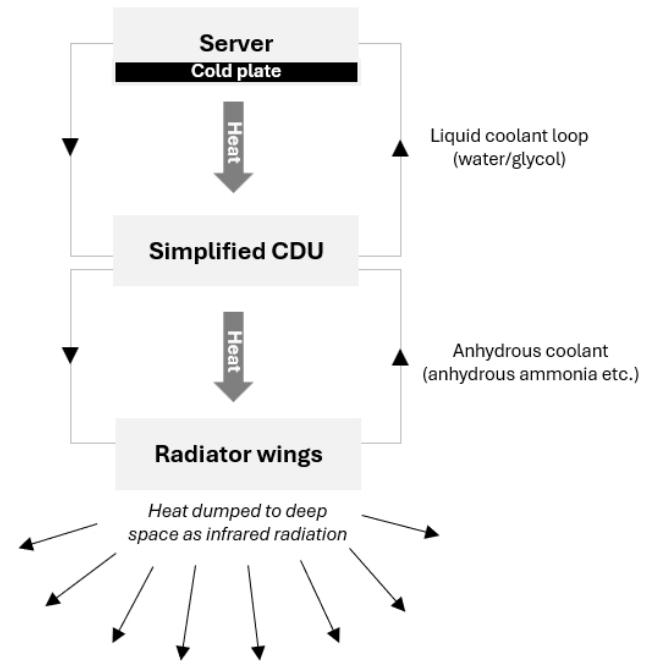
DETAILS

EXHIBIT 1: Liquid cooling architectures for terrestrial (land) and orbital (space) data centers

Terrestrial (land) data center



Orbital (space) data center



Source: Bernstein Analysis

EXHIBIT 2: Total cooling content opportunity by architecture (\$M/MW)

Component		Terrestrial	Orbital
Liquid cooling	Chillers ¹	~0.45 – 0.55	NA
	CRAH ²	~0.04 – 0.05	NA
	RDHx ²	~0.04 – 0.05	NA
	CDU ³	~0.30 – 0.40	~0.15 – 0.20
	Cold plates	~0.20 – 0.25	~0.20 – 0.25
	Manifolds	~0.05 – 0.10	~0.05 – 0.10
	Radiator wings	NA	Unclear
	DCIM software	~0.05 – 0.10	~0.05 – 0.10
Ancillaries (incl. refrigerant)		~0.10	~0.10
Redundancy allowance ⁴		~0.12 – 0.16	~0.06 – 0.08
Total Opportunity (\$M / MW)		~1.3 – 1.8	~0.6 – 0.8

↑
Excluding radiator wings

1. Fully loaded cost including chiller, installation labor, pipes, pumping, heat rejection equipment (dry coolers and cooling towers) 2. For terrestrial liquid-cooled: assuming ancillary heat capture split 50/50 among CRAH and RDHx 3. For terrestrial liquid-cooled: All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation; For orbital: assumed at 50% given simplification of equipment 4. Assumed 10%

Source: Bernstein Analysis

INTRODUCTION

In our view, data centers in space are best understood not as a fundamentally new form of computing infrastructure, but rather as a modified satellite (years-old demonstrated infrastructure) whose primary mission is computing AI workloads with embedded server racks.

Traditional satellites exist to perform a specific function (such as communications, navigation or scientific research). Orbital data centers simply substitute those payloads with GPU clusters optimized for AI workloads. As a result, many of the core architectural principles remain familiar to the aerospace industry. The platform requires a source of electrical power, a structural backbone, communications systems, thermal management equipment and the compute payload (i.e. server racks).

A useful way to think about the architecture is through three primary subsystems:

- Compute bus (server racks)
- Solar arrays
- Thermal management infrastructure

We are not discussing the mechanics of data centers in space in this note. Instead, this document focuses on the cooling infrastructure required for operating data centers in space. For additional information on orbital data center mechanics, please refer to our [initiation document on SpaceX](#).

LIQUID COOLING FOR DATA CENTERS IN SPACE

Recap of liquid cooling in terrestrial (land-based) data centers

Before discussing how cooling works in orbit, it is useful to recap how heat is removed from modern terrestrial data centers because orbital cooling is (in many aspects) simply a modified version of the same thermal management infrastructure. In traditional direct-to-chip liquid cooling, cold plates mounted directly on GPUs absorb heat and transfer it into a liquid coolant loop (water/glycol). Coolant distribution units (CDUs) regulate flow, pressure and temperature throughout the system while transferring heat into a facility-water loop. From there, chillers, cooling towers and dry coolers remove the heat from the facility and eject it into the surrounding environment. The core idea behind the setup is simple: transporting heat from the silicon chip to ambient air (outside the data center). This observation becomes critical when evaluating orbital cooling architectures because the ambient environment no longer exists in the traditional sense. In fact, something *better* exists: **cold, deep space**.

Our hypotheses on the mechanics of cooling in space

Regardless of whether a GPU sits inside a hyperscale facility in Northern Virginia or aboard a spacecraft ~800 kilometers above the Earth, every watt of electricity consumed ultimately becomes heat. The laws of thermodynamics remain the same. Orbital data centers simply utilize a different approach for managing the resulting thermal load. The governing physics are well understood and have been utilized for decades across spacecraft and satellites in general. The real question is not whether cooling in space is theoretically possible, but rather how the system scales when applied to multi-megawatt AI compute workloads.

A common misconception is that cooling should become easier because space is cold. While deep space certainly represents a large thermal sink, the absence of atmosphere introduces a different challenge: *how to eject this generated heat into space*. On Earth, heat can be removed through convection, where moving air or fluid carries thermal energy away (with the assistance of dry coolers / cooling towers). In orbit, convection disappears entirely because there is effectively no atmosphere, and thus *no medium to carry away the heat*. Heat cannot simply be blown away with fans or discharged into surrounding air (given there *is no air*). Instead, thermal energy must leave the system through radiation. We hypothesize that this is accomplished through “**radiator wings**” attached to the compute bus (large-scale modified versions of commonly used satellite radiators). Radiation thermodynamics is governed by the Stefan-Boltzmann law. The law states that every piece of matter emits thermal radiation, which is primarily a function of its temperature. In simple terms, the hotter a body is, the more radiation it emits, and vice versa. This is a fundamental property of matter, where all bodies with a temperature above absolute zero emit thermal radiation. This is the core underlying principle for the liquid cooling mechanics in orbital data centers.

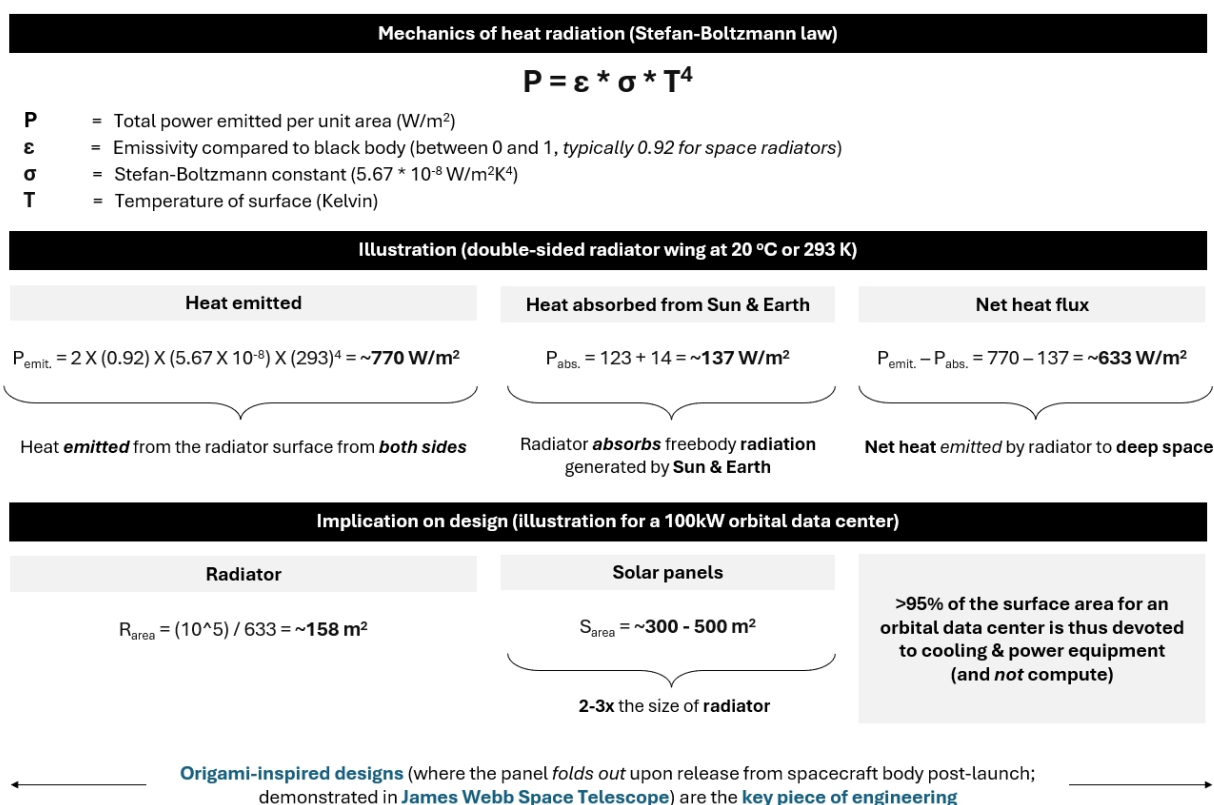
$$P = \epsilon * \sigma * T^4$$

where ϵ is the emissivity factor for the surface (0.92 for space radiators), σ is the Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$) and T is the temperature of the surface.

Using assumptions outlined in our analysis, a double-sided radiator (radiating heat from *both* sides) operating at approximately 20°C (or 293 Kelvin) can reject roughly ~633 W/m² of net heat (this takes into account the freebody radiation emitted by the Sun and Earth that is absorbed by any spacecraft in orbit). Applying this relationship to a ~100 kW compute platform implies ~158 square meters of radiator area, and ~300–500 m² of solar array area. Our analysis thus suggests that **>95% of the surface area** associated with an orbital data center may be devoted to cooling and power generation infrastructure rather than compute hardware itself. While investors typically focus on GPUs, servers and networking equipment, the dominant engineering challenge appears to be supporting those systems through sufficient solar power generation and thermal rejection capacity. These implications extend beyond cooling alone. Because all electrical power consumed by the compute cluster ultimately becomes heat, power generation and thermal management become linked design parameters. Larger compute deployments require larger solar arrays, which in turn support larger thermal loads that would invariably require larger radiator wings.

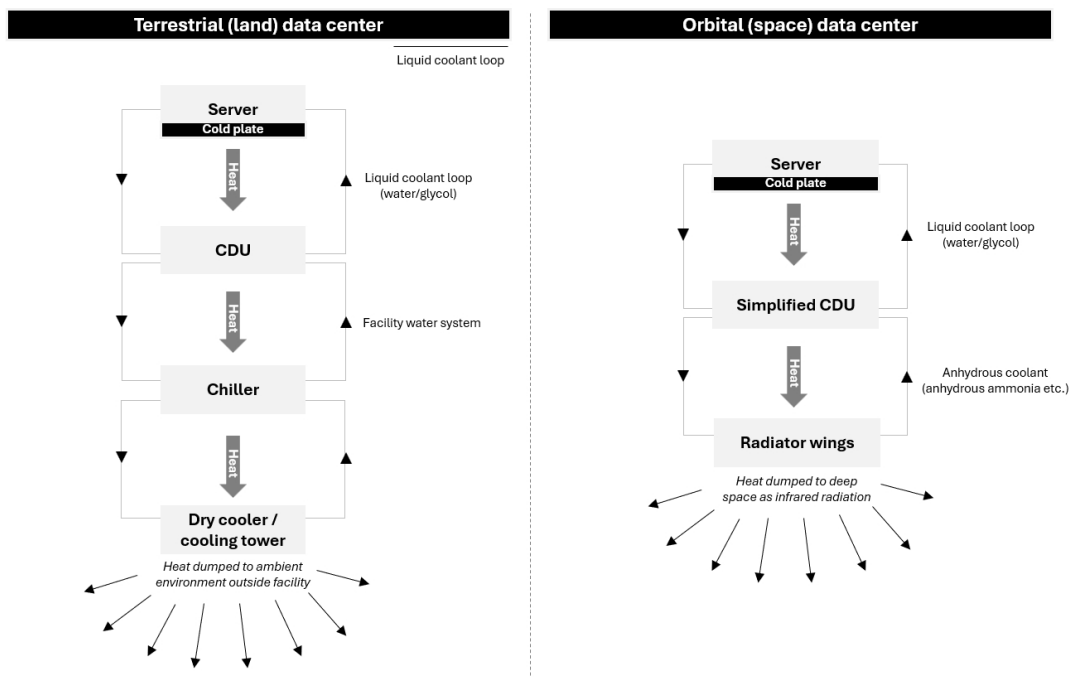
In simpler terms: an orbital data center is less a compute platform and more a power generation and heat rejection platform that just happens to contain compute hardware.

EXHIBIT 3: Core operating principle is simpler than we think



Source: Bernstein Analysis

EXHIBIT 4: Liquid cooling architectures for terrestrial (land) and orbital (space) data centers



Source: Bernstein Analysis

EXHIBIT 5: Our hypothesis on how orbital data center cooling works

How does orbital data center cooling work?

- Each orbital data center uses **direct-to-chip cold plates on GPU servers** (similar to traditional single-phase DTC), pulling heat into a closed liquid loop (e.g., water/glycol) that behaves similarly to terrestrial cold-plate systems but without air-side cooling
- The primary liquid loop (water/glycol mix) moves this waste heat to a **simplified CDU** (pump + heat exchanger), where heat is transferred into a **secondary loop** filled with **anhydrous coolant** such as anhydrous ammonia (long heritage of usage in spacecraft systems)
 - Anhydrous ammonia has high latent heat, **remains liquid across a wide temperature range in space** and is **compatible with aluminum tubing**
- The ammonia loop **circulates through the double-sided radiator wings**, which operate purely by radiation (*no convection in vacuum*), **emitting** heat in the form of **infrared energy** from both sides of the panel surfaces directly into **deep space**
- Radiator temperature, emissivity, and total surface area are co-optimized using the Stefan-Boltzmann law so that the **net radiative flux** (after accounting for absorbed Sun/Earth IR) matches the full IT load;
 - For a 100 kW orbital data center this implies $\sim 158 \text{ m}^2$ of radiator area and $\sim 400 \text{ m}^2$ of solar array area
- Because all compute power eventually becomes heat, **>95% of the spacecraft** surface area is allocated to **radiators and solar arrays**, with the cooling loop and power system dominating mass, deployment mechanics, and structural design rather than the GPU modules themselves
- Origami-style folding structures** (similar to James Webb space telescope's sunshield and radiators) enable radiator and solar-panel wings to **fit within launch vehicles**, then deploy and rigidize on orbit, maintaining stable geometry for the ammonia loop and consistent view factors in deep space

Source: Bernstein Analysis, Company Reports

Our hypothesis on how orbital data center cooling actually works

We hypothesize that the cooling architecture of an orbital data center is, in basic aspects, similar to that of a terrestrial liquid-cooled data center. The first half of the thermal management chain is effectively unchanged. Heat generated by GPUs is captured through direct-to-chip cold plates and transferred into a primary water/glycol coolant loop. The coolant absorbs heat from the GPUs and transports it away from the racks towards a coolant distribution unit (CDU). From the perspective of the server, cold plates, manifolds and primary liquid loop, an orbital data center looks remarkably similar to the direct-to-chip liquid-cooling architectures currently being deployed across hyperscale AI facilities.

The architecture begins to diverge once heat leaves the primary coolant loop. In a terrestrial facility, thermal energy is typically transferred into a facility water system (FWS) before ultimately reaching chillers, and then dry coolers. In orbit, the facility water system is effectively replaced by a secondary loop utilizing an anhydrous coolant (such as anhydrous ammonia which has a long history of use in spacecraft thermal management systems due to its favorable heat-transfer properties, broad liquid operating range, high latent heat and compatibility with lightweight aluminum tubing). This anhydrous coolant helps in transporting large quantities of thermal energy across long distances within the spacecraft while minimizing weight and pumping requirements, both of which are critical considerations in orbital applications.

The secondary loop ultimately circulates through large deployable radiator wings, which serve as the orbital equivalent of a cooling tower. Rather than rejecting heat into ambient air, the radiators emit thermal energy directly into deep space in the form of infrared radiation. As warm anhydrous fluid flows through the radiator panels, heat is continuously radiated from both sides of the structure, cooling the fluid before it returns to repeat the cycle. Consequently, the fundamental objective of the cooling system remains unchanged: transport heat from the chip to an external environment. The difference is simply the final heat sink. On Earth, heat is ejected to the ambient atmosphere through cooling towers and dry coolers; in orbit, heat is ejected to deep space through radiator wings via infrared radiation.

Our view on servicing and repairs

Unlike terrestrial data centers, where individual components can be replaced relatively easily (by a technician pulling the defective components out of the rack and physically replacing them), orbital data centers must assume that maintenance opportunities are **far more limited**. It is neither economically nor practically feasible to do a repair mission or extra-vehicular activity *just for servicing & repair*. Such missions usually cost multi-millions of dollars and require years-long planning per mission. So even if companies are able to sell equipment to orbital data centers, we do not believe there is a material service tail for them to monetize.

It is important to note that our hypothesis is grounded in the thermal management principles and cooling systems employed in current satellite and spacecraft designs. As the orbital data center ecosystem matures and industry participants provide greater technical transparency, we expect our understanding and analysis of these architectures to evolve accordingly.

U.S. MULTI-INDUSTRY & ELECTRICAL EQUIPMENT

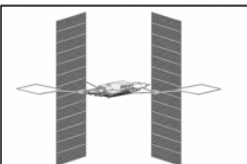
EXHIBIT 6: Components needed for liquid cooling (1/2)

Component	Description	Value (\$M/MW)	Requirement in architecture	
			Terrestrial	Orbital
Chiller ¹	 “Central engine” of the cooling plant, responsible for cooling water to inject into TCS and rejecting heat from warm liquid to the FWS	~0.45 – 0.55	✓ Required	✗ Not required
Dry cooler	 “Heat fans” to blow hot air outside of the facility	~0.10 – 0.20	✓ Required	✗ Not required
Cooling tower	 “Heat rejection equipment” sitting outside the facility to transfer heat from facility water system to environment	~0.10 – 0.20	✓ Required	✗ Not required
Computer room air handler (CRAH) ²	 Data-hall-level air handler that cools residual hot air over chilled-water coils	~0.40 – 0.60	✓ Required at partial capacity	✗ Not required

1. Fully loaded cost including chiller, installation labor, pipes, pumping, heat rejection equipment (dry coolers and cooling towers) 2. While gross unit price is shown, we have taken residual heat assumptions while calculating overall content cost for terrestrial liquid-cooled data center

Source: Bernstein Analysis

EXHIBIT 7: Components needed for liquid cooling (2/2)

Component	Description	Value (\$M/MW)	Requirement in architecture	
			Terrestrial	Orbital
Rear door heat exchanger (RDHx) ¹	 Server rack-level liquid-to-air coil that rejects heat exhaust from the back of the rack	~0.30 – 0.50	✓	✗
Coolant distribution unit (CDU) ²	 “Gateway” between facility water and chip-level liquid cooling loop; controls coolant flow, pressure and temperature to liquid-cooled racks	~0.30 – 0.40	✓	✓
Cold plates	 “Metal plates” mounted on GPUs / CPUs, responsible for pulling heat directly from chip into the liquid loop	~0.20 – 0.25	✓	✓
Manifolds	 Rack plumbing that distributes coolant from CDU to each cold plate and returns the warm liquid	~0.05 – 0.10	✓	✓
Radiator wings	 Deployable thermal panels that reject waste heat to deep space via infrared radiation	Unclear	✗	✓

1. While gross unit price is shown, we have taken residual heat assumptions while calculating overall content cost for terrestrial liquid-cooled data center 2. All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation. CDUs for orbital data centers are much smaller – consisting of a pump and heat exchanger (no facility-side refrigeration or condensation control needed)

Source: Bernstein Analysis

EXHIBIT 8: **Total cooling content opportunity by architecture (\$M/MW)**

Component		Terrestrial	Orbital
Liquid cooling	Chillers ¹	~0.45 – 0.55	NA
	CRAH ²	~0.04 – 0.05	NA
	RDHx ²	~0.04 – 0.05	NA
	CDU ³	~0.30 – 0.40	~0.15 – 0.20
	Cold plates	~0.20 – 0.25	~0.20 – 0.25
	Manifolds	~0.05 – 0.10	~0.05 – 0.10
	Radiator wings	NA	Unclear
	DCIM software	~0.05 – 0.10	~0.05 – 0.10
Ancillaries (incl. refrigerant)		~0.10	~0.10
Redundancy allowance ⁴		~0.12 – 0.16	~0.06 – 0.08
Total Opportunity (\$M / MW)		~1.3 – 1.8	~0.6 – 0.8



Excluding radiator wings

1. Fully loaded cost including chiller, installation labor, pipes, pumping, heat rejection equipment (dry coolers and cooling towers) 2. For terrestrial liquid-cooled: assuming ancillary heat capture split 50/50 among CRAH and RDHx 3. For terrestrial liquid-cooled: All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation; For orbital: assumed at 50% given simplification of equipment 4. Assumed 10%

Source: Bernstein Analysis

EXHIBIT 9: Recap | Terrestrial liquid-cooled content opportunity by company (\$M/MW)

Component									
Chillers + dry coolers	~0.45 – 0.55	<i>Seemingly not data-center grade</i>	~0.45 – 0.55	~0.45 – 0.55	~0.45 – 0.55	NA	NA	~0.45 – 0.55	NA
CRAH ¹	~0.04 – 0.05	NA	~0.04 – 0.05	~0.04 – 0.05	~0.04 – 0.05	NA	NA	~0.04 – 0.05	NA
RDHx ¹	~0.04 – 0.05	~0.04 – 0.05	NA	NA	NA	NA	NA	~0.04 – 0.05	~0.04 – 0.05
CDU ²	~0.30 – 0.40	~0.30 – 0.40	~0.30 – 0.40	~0.30 – 0.40	~0.30 – 0.40	~0.30 – 0.40	NA	~0.30 – 0.40	NA
Cold plates	~0.20 – 0.25	NA	NA	NA	NA	~0.20 – 0.25	NA	~0.20 – 0.25	NA
Manifolds	~0.05 – 0.10	~0.05 – 0.10	NA	NA	NA	~0.05 – 0.10	NA	~0.05 – 0.10	NA
Radiator wings	NA	NA	NA	NA	NA	NA	NA	NA	NA
DCIM software	~0.05 – 0.10	NA	NA	NA	NA	NA	NA	~0.05 – 0.10	NA
Overspec. allowance ³	~0.23 – 0.30	~0.08 – 0.11	~0.16 – 0.20	~0.16 – 0.20	~0.16 – 0.20	~0.11 – 0.15	NA	~0.23 – 0.30	~0.01 – 0.01
Total (\$M / MW)	~1.4 – 1.8	~0.5 – 0.7	~0.9 – 1.2	~0.9 – 1.2	~0.9 – 1.2	~0.7 – 0.9	NA	~1.4 – 1.8	~0.0 – 0.1

1. For terrestrial liquid-cooled: assuming ancillary heat capture split 50/50 among CRAH and RDHx 2. For terrestrial liquid-cooled: All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation; For orbital: assumed at 50% given simplification of equipment 3. Assumed 20% overspecification buffer

Forgent (FPS) not covered by Bernstein

Source: Bernstein Analysis

EXHIBIT 10: Orbital data center cooling content opportunity by company (\$M/MW)

Excluding radiator wings									
Component									
Chillers + dry coolers	NA	NA	NA	NA	NA	NA	NA	NA	NA
CRAH ¹	NA	NA	NA	NA	NA	NA	NA	NA	NA
RDHx ¹	NA	NA	NA	NA	NA	NA	NA	NA	NA
CDU ²	~0.15 – 0.20	~0.15 – 0.20	~0.15 – 0.20	~0.15 – 0.20	~0.15 – 0.20	~0.15 – 0.20	NA	~0.15 – 0.20	NA
Cold plates	~0.20 – 0.25	NA	NA	NA	NA	~0.20 – 0.25	NA	~0.20 – 0.25	NA
Manifolds	~0.05 – 0.10	~0.05 – 0.10	NA	NA	NA	~0.05 – 0.10	NA	~0.05 – 0.10	NA
Radiator wings	NA	NA	NA	NA	NA	NA	NA	NA	NA
DCIM software	~0.05 – 0.10	NA	NA	NA	NA	NA	NA	~0.05 – 0.10	NA
Overspec. allowance ³	~0.09 – 0.13	~0.04 – 0.06	~0.03 – 0.04	~0.03 – 0.04	~0.03 – 0.04	~0.08 – 0.11	NA	~0.09 – 0.13	NA
Total (\$M / MW)	~0.5 – 0.8	~0.2 – 0.4	~0.2	~0.2	~0.2	~0.5 – 0.7	NA	~0.5 – 0.8	NA

1. For terrestrial liquid-cooled: assuming ancillary heat capture split 50/50 among CRAH and RDHx 2. For terrestrial liquid-cooled: All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation; For orbital: assumed at 50% given simplification of equipment 3. Assumed 20% overspecification buffer

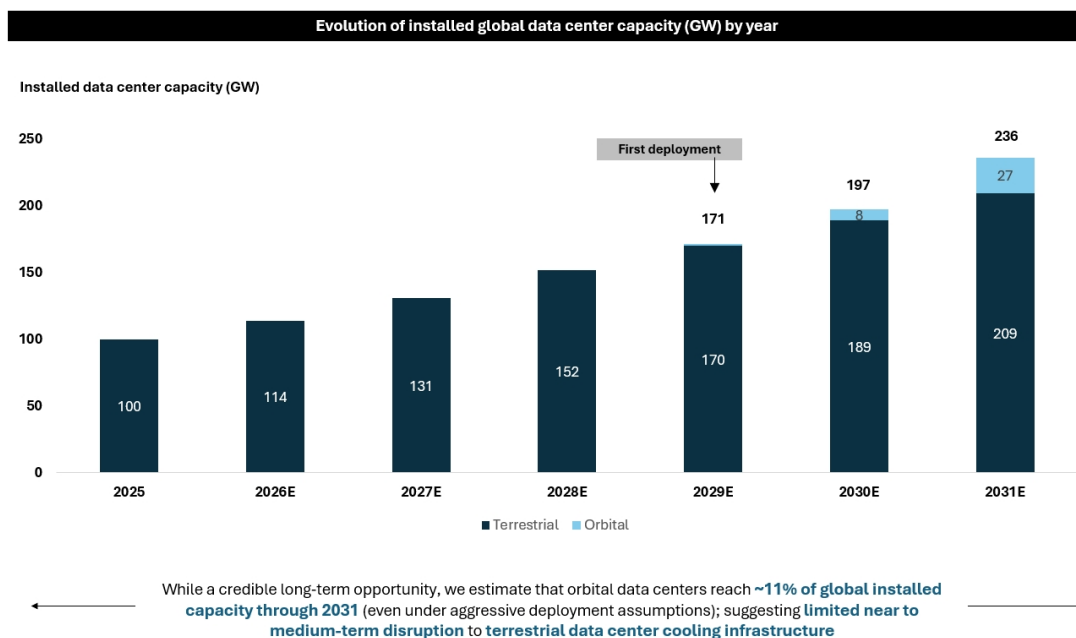
Source: Bernstein Analysis

LOOKING AHEAD

To assess the potential implications for cooling component manufacturers, we first mapped the cooling architecture for a terrestrial liquid-cooled data center and compared it against the hypothesized architecture for an orbital data center. We then evaluated the major components required under each configuration, before assigning a representative content value on a \$/MW basis. The exercise highlights that orbital data centers do not eliminate cooling infrastructure requirements, but rather *alter the composition* of the cooling stack. While traditional heat rejection equipment (such as chillers, cooling towers and dry coolers) largely disappear, cold plates, manifolds and CDUs remain essential, while radiator wings emerge as a new category. Basis our analysis, we estimate the total cooling equipment content opportunity at **\$0.6–0.8M/MW for orbital data centers (excluding radiator wings), versus \$1.3–1.8M/MW for terrestrial liquid-cooled data centers.**

That said, we would caution against extrapolating the orbital data center disruption too aggressively at this stage. While the concept is technically credible and a number of operators (led by SPCX) are actively pursuing commercial deployment, our analysis suggests the impact on the broader cooling ecosystem remains limited over the near-to-medium term. As outlined in [our initiation document on SpaceX](#), we estimate orbital data centers to account for ~27GW of total global installed capacity by 2031, representing ~1% of global installed data center capacity. Terrestrial infrastructure therefore remains the overwhelming majority of the market for the foreseeable future. As a result, our medium-term outlook for cooling component suppliers remains largely unchanged. The principal beneficiaries of AI-driven infrastructure growth continue to be companies exposed to terrestrial liquid cooling adoption, where deployment volumes are expected to remain significantly larger than those associated with orbital data centers.

Finally, while examining the space, we conducted a review of several new-age startups developing technologies specifically targeted at orbital compute infrastructure, including deployable radiator systems, integrated thermal management architectures, and other adjacent technologies. *While highly integrated orbital data center operators may ultimately choose to develop many of these capabilities in-house (SPCX has a strong bias for vertical integration),* we believe it is equally important to monitor the broader ecosystem of specialized suppliers who are in early stages of commercialization. For established thermal management and electrical equipment companies, these businesses may ultimately represent attractive acquisition or partnership opportunities should orbital data centers develop into a meaningful long-term market. Many data center cooling incumbents have used M&A to build exposure to liquid cooling over the last several years, we would not be surprised to see a similar playbook emerge around orbital thermal management technologies as the market continues to evolve.

EXHIBIT 11: **Orbital data centers remain a niche share of global capacity through 2031**

Source: IDC, Gartner, Omdia, Bernstein Global Data Center Model, Bernstein Analysis and Estimates

EXHIBIT 12: **While vertical integration remains a viable path, emerging cooling infrastructure providers for orbital data centers could be attractive acquisition targets**

Company						
Year founded	2003	~2000s	2024	2023	2023	2015
Country						
Funding / acquisition price	Undisclosed (acquired)	Undisclosed (acquired)	~\$200 million	~\$13.5 million	~\$10.1 million	~\$50 million
Key investors / acquirers	Blackstone	Phantom Space Corporation	Benchmark, EQT, Y Combinator	Alpha Funds, KDDI Green Partners	Kindred Ventures, Giant Step Capital	Razor's Edge, Lockheed Martin, Intel
Commentary	Two-phase heat pipes, composite radiator panels Logged 100M+ space flight hours; aggressively pivoting to orbital AI applications post Blackstone deal (2026)	Deployable radiator wings with high conductance hinges Now operates as Phantom Space subsidiary post acq. (Apr 2026); exclusively supplying the Phantom Cloud orbital data center constellation	"Stem-leaf" compute node design; developing radiator wings Flew the first orbital NVIDIA H100 GPU (Nov 2025); target completion of radiator wing-based cluster by Oct 2026	Proprietary TILE architecture (unifying solar array, compute, and radiator all-in-one) Each unit is a tile that layers solar cells, compute silicon and passive radiator into a "sandwich" structure; target commercial production by end of 2028	Space-rated radiation-hardened compute hardware for edge computing Flew the first NVIDIA Jetson GPU in orbit in 2025	Lightweight mesh solar cell arrays Positioned as a power enabler rather than a thermal hardware company

Source: Bernstein Analysis. All these companies are private companies.

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